Probability reference

Bernoulli and binomial

An observation $X \in \{0, 1\}$ has probability $P(X = 1) = p$. The expectation $\mathbb{E}[X] = p$ remains beside selectable prose, with variance $\operatorname{Var}(X) = p(1-p)$.

$$P(X = k) = p^k (1-p)^{1-k}, \quad \mathbb{E}[X] = p, \quad \operatorname{Var}(X) = p(1-p)$$

An inline fraction $\frac{r(1-p)}{p^2}$ retains its numerator and denominator. This complete final sentence remains searchable.

Imported notation $P(Y = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad \mathbb{E}[Y] = np.$

Imported units $10\ \text{minutes}$ and $49\ \text{mm}$ remain attached to their values.

Imported expectation $\mathbb{E}[A + T] = 20\ \text{minutes}$ stays in one expression.

Matrices and systems

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

$$f(x) = \begin{cases} x^2 & x > 0 \\ 0 & x \le 0 \end{cases}$$

Math fence

$$I = \frac{\text{observed events}}{\text{total events}} \times 100$$

Wide comparison

Model	Mass	Mean	Variance	Context
Binomial	$\binom{n}{k} p^k (1-p)^{n-k}$	np	$np(1-p)$	Independent observations in a fixed sample.
Poisson	$\frac{\lambda^k e^{-\lambda}}{k!}$	λ	λ	Events counted during a fixed interval.
Geometric	$(1-p)^{k-1} p$	$\frac{1}{p}$	$\frac{1-p}{p^2}$	Attempts until the first event.

Final portrait section

All five table columns and this final paragraph must remain in the PDF. Literal code remains literal.

```
print("\\Delta is literal code")
```